

KARTIK BALASAHEB KAPSE

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PROFILE SUMMARY

- Master's student in AI Engineering for Autonomous Systems at Technische Hochschule Ingolstadt, specializing in Generative AI, Reinforcement Learning, Machine Learning, and Deep Learning
- Proficient in Python, PyTorch, Scikit-learn, MATLAB, Simulink, and model-based system development for safety-critical applications
- Passionate about leveraging deep tech-including LLMs, HPC, data-driven modeling, and prototype testing to drive innovative product development and support sustainable engineering solutions

EDUCATION

03/2024 to Present	M.Eng (AI Engineering of Autonomous Systems) Technische Hochschule Ingolstadt, Germany Noteworthy modules: AI and automotive system, Principle of Autonomy & decision-making (Reinforcement Learning), Systems Engineering & Architecting for Edge Computing
07/2018 to 05/2022	B.Eng (Electrical Engineering) Savitribai Phule Pune University, Pune, India Noteworthy modules: Advanced microcontrollers and applications, Power system operations and control, and electrical machines Thesis: VFTO Analysis in 245kV Gas-Insulated Substations

WORK EXPERIENCE

05/2026 to Present	Master Thesis Student NETZSCH Pumpen & Systeme GmbH, Waldkraiburg, Deutschland • Development of a Self-Supervised and Edge-Deployable Degradation Monitoring System for Positive Displacement Pumps Using Vibration Data Under Varying Operating Conditions
10/2025 to 04/2026	Working Student (Research & Development - ThinkIsense) NETZSCH Pumpen & Systeme GmbH, Waldkraiburg, Deutschland • Built Python based data pipelines for time and frequency domain signal analysis and feature engineering • Designed, trained, and evaluated ML/DL models using PyTorch, and scikit-learn • Developed AI algorithms for anomaly detection and performance prediction
04/2025 to 03/2026	Student Assistant (Research & Development - FEDIMA) Technische Hochschule Ingolstadt (AIMotion Bavaria), Deutschland • Modular Neural Networks for Quality Prediction in Manufacturing • Conducted research on modular neural network, ModFL/FedPer, Domain Adversarial Neural Networks architectures to enable adaptive ML-based quality prediction in manufacturing systems • Prepared and preprocessed tabular datasets collected from heterogeneous production lines with varying measurement inputs • Implemented and evaluated various modular network architectures using Python and PyTorch • Gained hands-on experience in feature engineering, data modeling, and scalable ML design for industrial applications
01/2023 to 08/2023	Senior Executive Operations (SCADA Distribution System) Adani Electricity Mumbai Limited, India • Led SCADA-based outage management and real-time system optimization • Developed process automation scripts and collaboration on predictive maintenance models
06/2022 to 01/2023	Graduate Engineer Trainee (Operations & Maintenance) Adani Electricity Mumbai Limited, India

08/2019 to 01/2022

- Involved in switchgear operations, system monitoring, fault detection, and analysis
- Built foundational experience in electrical system modelling

LV System Lead & Battery Pack Design Engineer

Team Brahmastra - Formula Student, India (Work-Study Member)

- Designed sensor integration and safety system inputs for the shutdown circuit in the Low Voltage (LV) system
 - Secured *5th rank in FMEA* out of 37 teams in Formula Bharat
- Designed a lithium-ion battery pack using LiFePO₄ 32650 cells for the High Voltage (HV) system
 - Ranked *10th in Overall Assessment* in Formula Green by ISNEE

SKILL-SET

- **Programming & Tools:** Object-Oriented Programming Python (Advanced), C/C++ (Intermediate), SQL (Basic), MATLAB (Intermediate), Linux (Basic), Git- version control(Advanced), Microsoft Office: Excel, PowerPoint, Word(Advanced), Thingsboard
- **AI & ML Tools:** PyTorch, scikit-learn, TensorFlow, Keras, Numpy, Pandas, HuggingFace , LangChain, Gradio, ChromaDB
- **Modeling & Simulation:** Simulink, Simscape
- **Data Science:** EDA, Statistics, Analytics, Predictive Modeling, Visualization, NLP, Image Processing, MySQL, RAG, GraphRAG
- **Automation & Debugging:** Pipelines, Automation scripts in MATLAB/Python, Model Validation & Testing, Test Automation
- **Cloud Computing:** Familiarity with AWS, Azure DevOps
- **Soft Skills:** Proactive, Problem-Solving, organized, Communication, Analytical

LANGUAGES

- **English:** C1 (IELTS) - Fluent - spoken and written
- **Deutsch:** A2 (Goethe-Zertifikat) Basic - spoken and written
- **Marathi, Hindi:** Native/Bilingual proficiency

PROJECTS & AI RESEARCH

- RAGMentor: AI-Powered Technical Learning Assistant
- End-to-End Learning vs. Modular Architectures: Comparative study of autonomous driving systems
- Developing Autonomous Driving Agent using PPO: Reinforcement learning for lane-keeping and obstacle avoidance in MetaDrive
- Digit Recognition Using CNN/MLP: Deep learning models using Hugging Face and PyTorch on MNIST
- Q-learning for Parachute Lander Simulation: Autonomous control using discrete action space
- Steel Surface Defect Classification: GridCV, PCA, and Scikit-learn on Kaggle dataset
- VFTO Analysis in Gas-Insulated Substations: JRAEE Journal Vol.5

CERTIFICATES

- **NVIDIA DL Institute:** Building LLM Applications With Prompt Engineering, Fundamentals of Accelerated Data Science
- **MathWorks Academy:** MATLAB Fundamentals, Deep Learning Toolbox, Machine Learning, Simulink & Embedded Coder
- **Coursera:** Getting Started with Python, Electric Power System, Introduction to Programming with MATLAB
- **IBM:** Data Analysis Using Python, Python for Data Science
- **Udemy:** Open-Source Drone Development, Machine Learning A-Z: AI, LLM Engineering: Master AI, Large Language Models & Agents
- **IEEE Workshop:** Electric Vehicle Design using MATLAB
- **Participation:** Formula Green ISNEE Motorsport, FSEV Concept Challenge, Entrepreneurship Convention at Vishwapreneur19

Munich, 13.06.2026

